

Assignment

Maintenance Modelling and Analysis

Deadline: **Thursday June 18th**

Describe in detail all steps you take, the choices you make, the model you use, and the results in a comprehensive report. Use figures and tables to present your results and support your conclusions. **The word limit in the report is 5000 words (This is quite a lot, so less is no problem!)**

The report shall not contain any computer code. Add your Python code in BrightSpace as a zip file. Make sure that it is clear and well commented, and that it works.

We expect the report to be clear and written in an understandable way. Though we do not give any specific points for this, we might subtract points if this is not the case.

Submit your report and model script file(s) through BrightSpace at the latest on Thursday June 18th. Don't forget to include the group number, names, and student IDs in the report and script file(s). Files submitted by email will not be considered. If you fail to meet the deadline, 0.5 points will be deducted from your grade for each day past the deadline.

Minimum grade: If you fail to obtain a grade of 5.5 or higher you will fail the assignment. In that case, you will get a chance to improve your work and pass the assignment. Your final grade cannot become higher than 5.5 in that case.

Grading of the report is: See assignment.

Groups: The assignment is meant for groups of two. If you want to form a group of three students instead, please come to me so that I can extend the assignment.

Self-enrollment: Self-enroll your group in Brightspace. Also hand-in the assignment with this group.

Generative AI: At the end of your report, shortly explain if, and if so, how you used AI tools (chatGPT etc.) in this assignment. No points will be deducted for this, but we will use it to enhance the quality of the assignment in later years.

Work distribution At the end of the report, please add a small table with the workload distribution for your group. If you all agree that everyone in the group contributed equally, you can simply state "All members of the group contributed equally" and omit this table.

Name	Part 1		Part 2.1		Part 2.2		Part 2.3		Part 2.4	
	Code	Report	Code	Report	Code	Report	Code	Report	Code	Report
#1	#%	#%	#%	#%	#%	#%	#%	#%	#%	#%
#2	#%	#%	#%	#%	#%	#%	#%	#%	#%	#%

Data and packages

You are going to i) optimize a preventive maintenance strategy (part 1) and ii) develop a Remaining Useful Life (RUL) prediction model (part 2) for aircraft turbofan engines. For this, we will use the C-MAPSS data, which can be found at: <https://www.nasa.gov/intelligent-systems-division/discovery-and-systems-health/pcoe/pcoe-data-set-repository/> (6th dataset in this link). The dataset consists of four subsets. We will only look at dataset FD001. This dataset contains for each engine a certain number of flight cycles (cycle 1: First cycle it has been used. Last cycle: Cycle during which it failed), and the sensor measurements of each flight cycle.

I have provided some code to import this dataset on Brightspace. In this way, you will ensure that you have the right variable names.

An useful package for part 1 of the assignment is reliability, see <https://reliability.readthedocs.io/en/latest/>.

Part 1: Preventive maintenance (25/100 points)

The first part focuses on preventive maintenance. For this question, assume that the dataset FD001 does *not* contain any sensor measurements or operating conditions. Instead, you only know for each engine how many flight cycles it has been used.

There is no need to give analytical derivations for this assignment. You can calculate all answer numerically, including the hazard function and optimal maintenance time, and use packages to fit the distributions. Do make sure to use the correct mathematical notation.

Assume that preventive maintenance on on aircraft engine costs 10,000 euros, while corrective maintenance costs 100,000 euros. Consider an age-based preventive maintenance policy, where we replace engines preventively after t flight cycles, or upon failure (whatever is first). Using the **train** set of FD001 to optimize this maintenance strategy, as explained in lectures 3-5. Include the following aspects:

- Fit all, or multiple, distributions available in the reliability package to the data. Which distribution(s) provide the best fit? Is this in line with your expectation? Select one (or more) distribution(s) to continue this exercise with, and motivate why you selected this distribution/these distributions. Take into account several ways to analyse the best fit, including a visualization. (12 points)
- Visualize and describe the shape of the hazard function(s) of the best fitted distribution(s) (there is **no** need to provide the analytical formula). What does the shape tell you about the possible optimal maintenance strategy? (5 points)
- Calculate the optimal value of t and the corresponding long-term average maintenance costs $g(t)$. Visualize your answer (8 points)

Part 2: Predictive maintenance (75/100 points)

This second part focuses on predictive maintenance. For this part, you may use all sensor measurements and the measurement on the operating conditions available for subset FD001. The goal is to develop a machine learning model that predicts the RUL of an aircraft engine, given the measurements belonging to this engine. Points are given, among others, for the correct implementation and explanation of the chosen model, the final results, how advanced the model is, the motivation of why this model is suitable for RUL prediction, advanced techniques (for instance for explainability, to prevent overfitting, and/or for hyperparameter tuning), the possible comparison of multiple models, the description of the model in the report, etc.

For the test set, the RUL values of each engine at the last (maximum) cycle are in a separate file. You only need to predict the RUL of the test engine at this last cycle, i.e., I expect that you make 100 different RUL predictions in total (corresponding to 100 test engines).

When an engine is still healthy, it is very difficult to accurately predict the RUL. For this dataset, the RUL is therefore usually capped at 125, i.e., if the RUL is larger than 125 flight cycles, we aim to predict 125 flight cycles regardless, for both the training and the test set. A RUL prediction close to 125 flight cycles can be seen as an indication that the engine is healthy. Feel free to also make this cap: It will help you to obtain the desired model accuracy.

If you have not followed any machine learning course, a relatively easy machine learning method are decision trees/random forests/boosting algorithms for decision trees. Information about this can be found in the “Old slides” map on Brightspace, and in Chapter 8 of the book “An introduction to statistical learning with applications in R: By Gareth James, Daniela Witten, Trevor Hastie, and Robert Tibshirani, New York, Springer Science and Business Media, 2013”. However, also without a machine learning course, you can of course select more complicated machine learning algorithms as well.

Describe the data, your machine learning model, how you trained the model and the results. Pay specific attention to:

1. Describe the data in this dataset. Based on this initial data exploration, are there any sensors you would disregard? (10 points)
2. Describe the machine learning model(s) that you used for RUL prognostics. Clearly describe the model and all steps you take in training the model (including any data preprocessing steps) using the appropriate mathematical notation. Also explain why you choose this model(s) for RUL prediction. (30 points)
3. Describe and explain the hyperparameter tuning of your model(s). Make sure the different engines are not mixed in your training, validation and test split (10 points).
4. Give a clear description of your results on the test set, using both figures and tables. I expect that at least one final model will have a RMSE of at most 20 flight cycles on the test set (25 points).